

Master Systematic Review Matrix: Study Characteristics, Treatments, and Predictors of END

	n	Age	Baseline NIHSS	LVO	IVT	Primary EVT	Rescue EVT	END Def.	END %	Clinical (Uni)	Clinical (Multi)	Radiological (Uni)	Radiological (Multi)
Boulenoir et al. 2021	74	Mean: 64.0	Median: 3.0	BAS, ICA, MCA	Yes (100.0%)	No	Yes (54.0%)	≥4 pts / within 24h	29.7%	Diabetes mellitus absent, Non-atheromatous etiology	None	Intracranial occlusion post-IVT, Suprabulbar iICAo	Suprabulbar iICAo
Boulenoir et al. 2022	189	Mean: 65.0	Median: 3.0	ICA	Yes (50.0%)	No	Yes (54.0%)	≥4 pts / within 24h	24.0%	None	IVT, Non-atheromatous etiology	None	None
Broccolini et al. 2024	208	Median: 70.0	Median: 3.0	MCA	Yes (64.0%)	No	Yes (100.0%)	≥4 pts / within 24h	41.8%	AFib, Current anticoagulant therapy, Current antiplatelet therapy	AFib	Dominant M2	None
Gwak et al. 2021	94	Mean: 66.9	Mean: 2.12	ICA, MCA	Yes (29.8%)	No	Yes (62.2%)	≥2 pts / within 72h	39.4%	IVT, Shorter onset-to-door time	IVT	Higher Tmax>6s volume, Mismatch vol ≥32.77 mL (Tmax >6s – DWI), Tmax 4-6s vol ≥55.20 mL (mild hypoperfusion), Tmax >6s vol ≥53.73 mL (critical hypoperfusion)	Mismatch vol ≥32.77 mL (Tmax >6s – DWI), Tmax 4-6s vol ≥55.20 mL (mild hypoperfusion), Tmax >6s vol ≥53.73 mL (critical hypoperfusion)
Gwak et al. 2022	94	Mean: 66.9	Mean: 2.12	ICA, MCA	Yes (29.8%)	No	Yes (24.5%)	≥2 pts / within 72h	39.4%	IVT, Shorter onset-to-door time	IVT	Poor perfusion profile	Poor perfusion profile
He et al. 2021	134	Mean: 67.01	Median: 2.0	ICA, MCA	Yes (26.1%)	No	No	≥4 pts / within 24h	20.2%	Higher baseline glucose	Higher baseline glucose	Longer SVS, SVS ≥9.45 mm	Longer SVS, SVS ≥9.45 mm
Heitkamp et al. 2025	817	Median: 72.0	Median: 4.0	ICA, MCA	Yes (41.0%)	Yes	No	≥4 pts / within 24h	23.5%	Higher pre-stroke mRS, Lower baseline NIHSS, No Afib, No IVT	Higher pre-stroke mRS, Lower baseline NIHSS	Adverse events during EVT, General anesthesia, Higher no of passes, Intracranial ICA, Longer admission-to-groin-puncture time, Unsuccessful recanalisation (TICI<2b)	Adverse events during EVT, General anesthesia, Higher no of passes, ICH within 24h, Longer admission-to-groin-puncture time
Hu et al. 2025	163	Mean: 66.3	Median: 2.0	BAS, ICA, MCA, VA	Yes (100.0%)	No	Yes	≥4 pts / within 24h	22.1%	Age, Delayed antiplatelet therapy	Delayed antiplatelet therapy	Isolated ICA occlusion, Tmax >4s volume	Isolated ICA occlusion
Itabashi et al. 2023	35	Median: 73.0	Median: 2.0	ICA	Yes (11.4%)	No	Yes (8.6%)	≥1 pts / Hosp.	37.0%	Diabetes mellitus present, EVT, Higher baseline NIHSS, IVT	no multivariable analysis done, none	None	No multivariable analysis done
Kim et al. 2022	1083	Mean: 67.2	Median: 2.0	ICA, MCA	Yes (10.8%)	Yes	Yes (11.1%)	≥2 pts / Hosp.	19.1%	Higher baseline NIHSS, Hypertension	Higher baseline NIHSS, Hypertension	Extracranial ICA occlusion	Extracranial ICA occlusion, Tandem occlusion
Lee et al. 2020	81	Mean: 65.8	Mean: 4.3	ICAT, MCA	Yes (100.0%)	No	Yes (44.0%)	≥4 pts / within 24h	34.6%	Diabetes mellitus present, Older age	no multivariable analysis done, none	Higher Tmax>6s volume, Larger CTP core, Proximal LVO: M1/ICAT	No multivariable analysis done, none
Mazya et al. 2018	2336	Median: 67.0	Median: 4.0	ACA, BAS, ICA, ICAT, MCA, PCA, VA	Yes (100.0%)	No	No	≥4 pts / within 24h	9.0%	Heart failure	no multivariable analysis done, none	None	None
Min et al. 2020	166	Mean: 66.0	Median: 2.0	ICA, MCA	Unclear	No	Yes	≥1 pts / Hosp.	38.6%	None	None	None	Insular lesions
Saleem et al. 2020	122	Mean: 62.0	Median: 3.0	ACA, BAS, ICA, MCA, PCA	Yes (25.4%)	No	Yes (54.2%)	≥4 pts / Hosp.	19.7%	None	None	None	None
Seners et al. 2021	1076	Mean: 70.0	Median: 3.0	BAS, ICA, MCA	Yes (100.0%)	No	Yes (56.0%)	≥4 pts / within 24h	12.1%	Male sex	None	Longer thrombus, More proximal occlusion	Longer thrombus, More proximal occlusion, Thrombus>9mm
Shi et al. 2025	347	Mean: 66.9	Median: 2.0	BAS, ICA, MCA, VA	Yes (44.7%)	No	Yes (100.0%)	≥4 pts / within 24h	12.1%	Higher SBP variability, Higher admission SBP	Admission SBP > 150mmHg, SBP variability >24 mmHg, highest quartile of 24h SBP variability, highest quartile of admission SBP	Higher Tmax>6s volume	Tmax>6s volume - highest quartile
Wang et al. 2022	39	Mean: 64.23	Median: 3.0	ACA, ICA, MCA	No	No	Yes	≥4 pts / within 24h	48.7%	NFL	NFL, NFL >55.03 pg/ml	None	None